

Solve problems involving the area and perimeter of irregular and composite shapes using appropriate units

Learning intention

- solve problems involving the area and perimeter of irregular and composite shapes using appropriate units.

Teach and model

- determining the area of composite shapes by composing or decomposing shapes.
- determining the perimeter and area of irregular shapes by sums of increasingly accurate covering measurements, such as line segments and grids.
- using arrays and rectangles to approximate the area of irregular shapes in situations such as a council needing to work out how much mosquito spray to use for a swamp area or a farmer needing to work out how much seed.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.